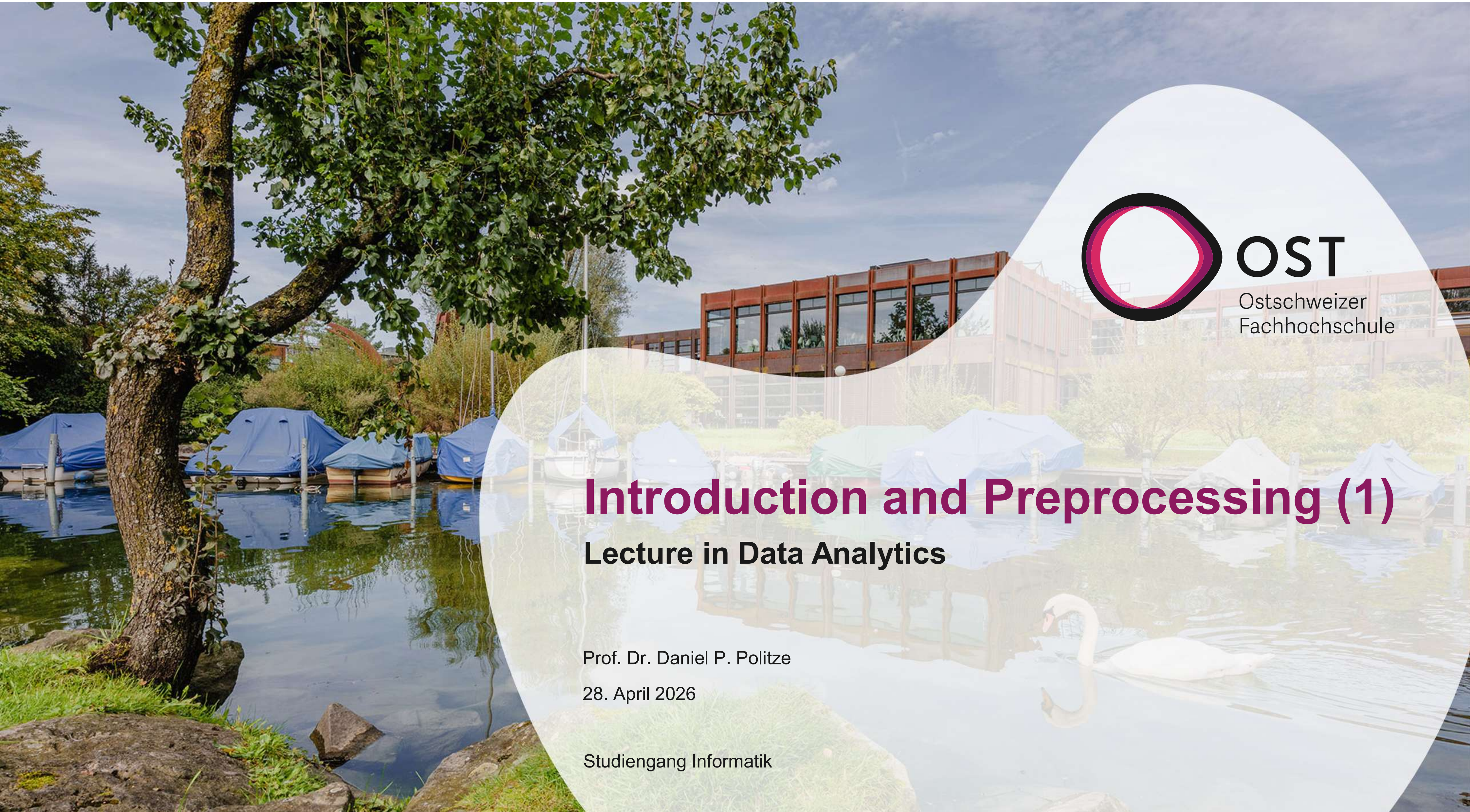

OST
Ostschweizer Fachhochschule

# Introduction and Preprocessing (1)

## Lecture in Data Analytics

Prof. Dr. Daniel P. Politze

28. April 2026

Studiengang Informatik

# Disclaimer

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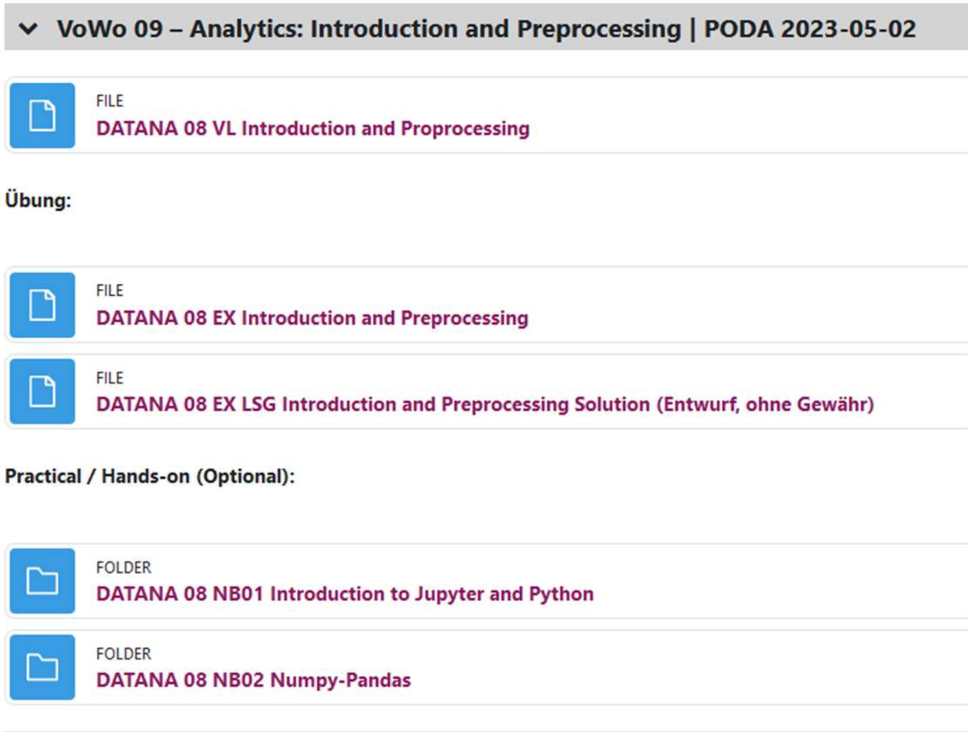
## Agenda

# What we will cover today…

- **Administratives / Course Overview**
- What is Data Science?
- What kind of patterns can be mined?
- What is Data like?
- Data Quality and Preprocessing Motivation
- Basic Statistical Description of Data
- Rough Data Visualization Overview

# Organizational Issues & Grading

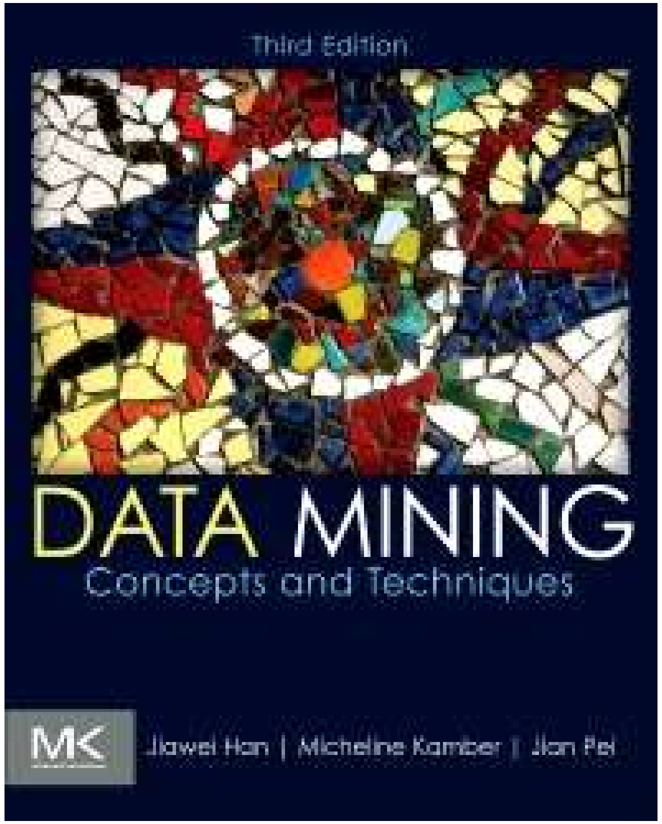
- Have a look at Moodle
  - Lecture Slides will be found here
  - Exercises will be found here
  - Links and additional literature will be found here
  - Important announcements will be found here
- Lecture Stream on SwitchTube (2wks)
- Final exam
  - 60 minutes (of total 120 minutes)
  - covering all the theory and exercises
  - there is no semester project
    but an optional Jupyther-Notebooks to solve



# Course Book

- Data Mining: Concepts and Techniques
- (3rd Edition)

- Authors: Jiawei Han Micheline Kamber Jian Pei
- Hardcover ISBN: 9780123814791
- eBook ISBN: 9780123814807
- Imprint: Morgan Kaufmann
- Published Date: 22nd June 2011
- Page Count: 744



# Course Syllabus «Data Science»

| | | |
|---|---|---|
| **1** | **Introduction** | **Chapter 1** |
| | **Preprocessing & Visualization – Part 1 (data objects, basic statistics)** | **Chapter 2** |
| 2 | Preprocessing & Visualization – Part 2 (data cleaning, correlation) | Chapter 3 |
| | Preprocessing & Visualization – Part 3 (data transformation, discretization) | Chapter 3 |
| 3 | Association Analysis | Chapter 6 |
| | Classification – Part 1 (basic concept, naive bayes, evaluation) | Chapter 8 |
| 4 | Classification – Part 2 (decision trees) | Chapter 8 |
| | Classification – Part 3 (ensemble methods) | Chapter 8 |
| 5 | Clustering – Part 1 (basic concept, partitioning methods, measuring similarity) | Chapter 10 |
| | Clustering – Part 2 (hierarchical methods) | Chapter 10 |
| 6 | Clustering – Part 3 (density-based methods) | Chapter 10 |
| | Clustering – Part 4 (outlook on advanced methods, evaluation) | Chapter 10 |

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# Early Evidence – Kepler the Data Scientist

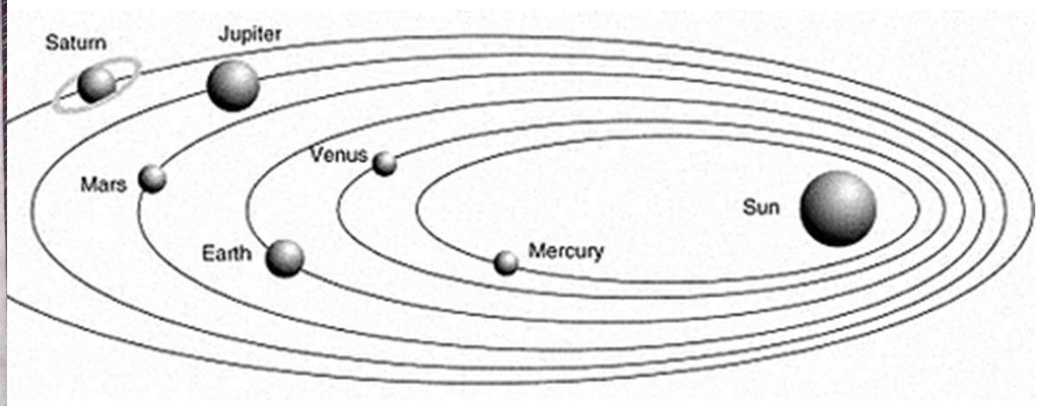
- Used the Rudolphine Tables
  (Tabulae Rudolphinae)
- Performed analysis of data collected by Tycho Brahe
- Discovered elliptical orbits of the planets (heliocentric model)

Johannes Kepler
(1571 - 1630)

TABULÆ
RUDOLPHINÆ,
QUIBUS ASTRONOMICÆ SCIENTIÆ, TEMPOrum longinquitate collapsæ RESTAURATIO continetur;
A Phœnice illo Astronomorum
TYCHONE,
Ex Illustri & Generosa BRAHEORUM in Regno Daniæ familiâ oriundo Equite,
PRIMUM ANIMO CONCEPTA ET DESTINATA ANNO CHRISTI MDLXIV: EXINDE OBSERVATIONIBUS SIDERUM ACCURATISSIMIS, POST ANNUM PRÆCIPUE MDLXXII, QUO SIDUS IN CASSIOPEIÆ CONSTELLATIONE NOVUM EFFULSIT, SERIÒ AFFECTATA; variisque operibus, cùm mechanicis, tùm librariis, impenso patrimonio amplissimo, accedentibus etiam subsidiis FRIDERICI II. DANIÆ REGIS, regali magnificentiâ dignis, tracta per annos XXV. potissimùm in Insula freti SUNDICI HUENNA, & arce URANIBURGO, in hos usus à fundamentis extructâ:
TANDEM TRADUCTA IN GERMANIAM, INQUE AULAM ET Nomen RUDOLPHI IMP. anno MDIIC.

TABULAS IPSAS, JAM ET NUNCUPATAS, ET AFFECTAS, SED MORTE AUTHORIS SUI ANNO MDCI. DESERTAS,
JUSSU ET STIPENDIIS FRETUS TRIUM IMPPP.
RUDOLPHI, MATTHIÆ, FERDINANDI,
ANNITENTIBUS HÆREDIBUS BRAHEANIS; EX FUNDAMENTIS OBservationum relictarum; ad exemplum ferè partium jam exstructarum; continuis multorum annorum speculationibus, & computationibus, primùm PRAGÆ Bohemorum continuavit; deinde LINCII, Superioris Austriæ Metropoli, subsidiis etiam Ill. Provincialium adjutus, perfecit, absolvit; adq; causarum & calculi perennis formulam traduxit.
IOANNES KEPLERUS,
TYCHONI primùm à RUDOLPHO II. Imp. adjunctus calculi minister; indèq; Trium ordine Imppp. Mathematicus:
Qui idem de speciali mandato FERDINANDI II. IMP. petentibus instantibúsq; Hæredibus,
Opus hoc ad usus præsentium & posteritatis, typis, numericis propriis, cæteris, & prælo JONÆ SAURII, Reip. Ulmanæ Typographi, in publicum extulit, & Typographicis operis ULMÆ curator affuit.

Cum Privilegiis, IMP. & Regum Rerúmq; publ. vivo TYCHONI ejúsq; Hæredibus, & speciali Imperatorio, ipsi KEPLERO concesso, ad Annos XXX.
ANNO M. DC. XXVII.







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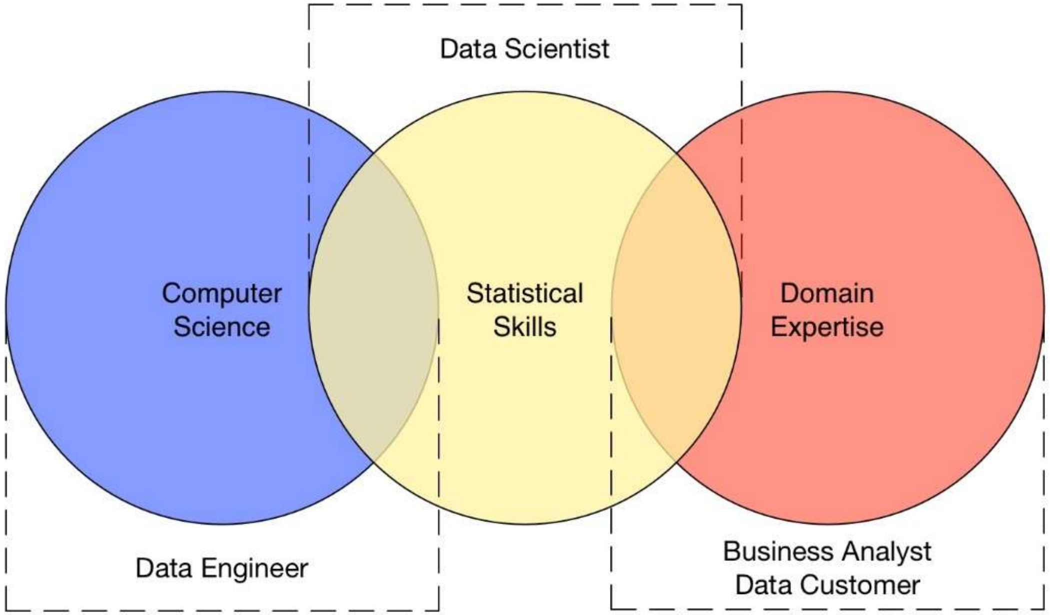
# What is Data Science about?

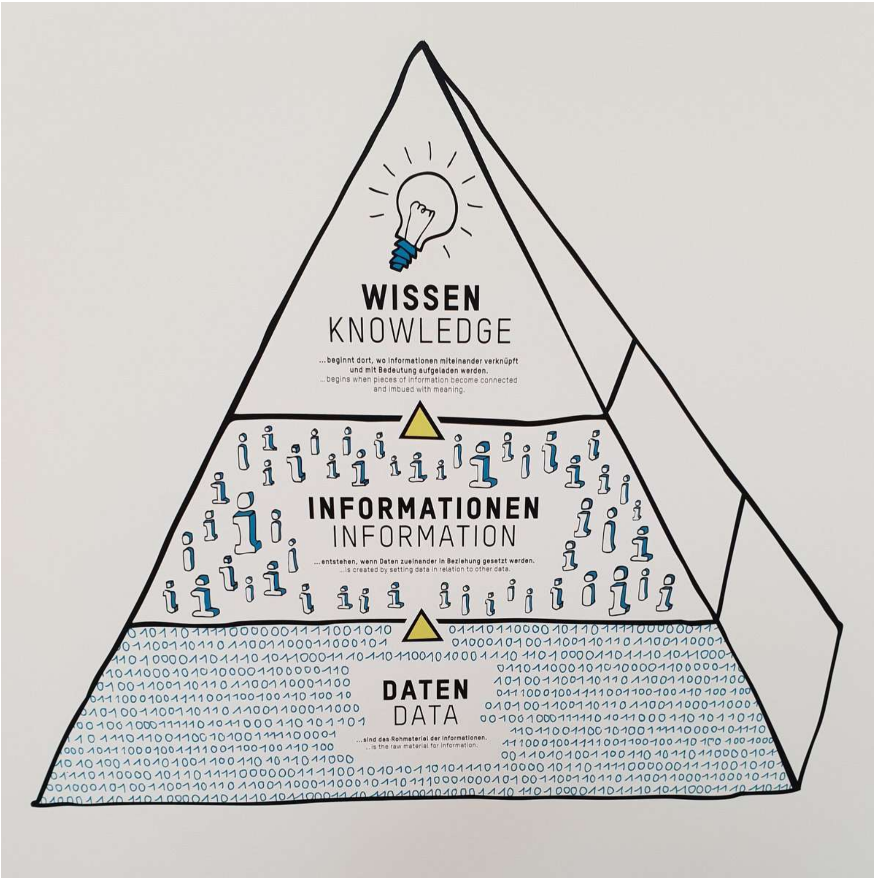
- It’s about extracting knowledge from data
  - Sometimes from a lot of data (Big Data)
  - Discover new (unknown) and useful patterns th




- It’s interdisciplinary
  - Business needs must be understood
  - **Uses Data Mining methods** for the extraction
  - Hypotheses must be formulated and tested
  - Models must be evaluated
  - Results must be presented to the audience in the right way
  - Conclusions must be understood und correct measures must be taken

## What is Data Science?

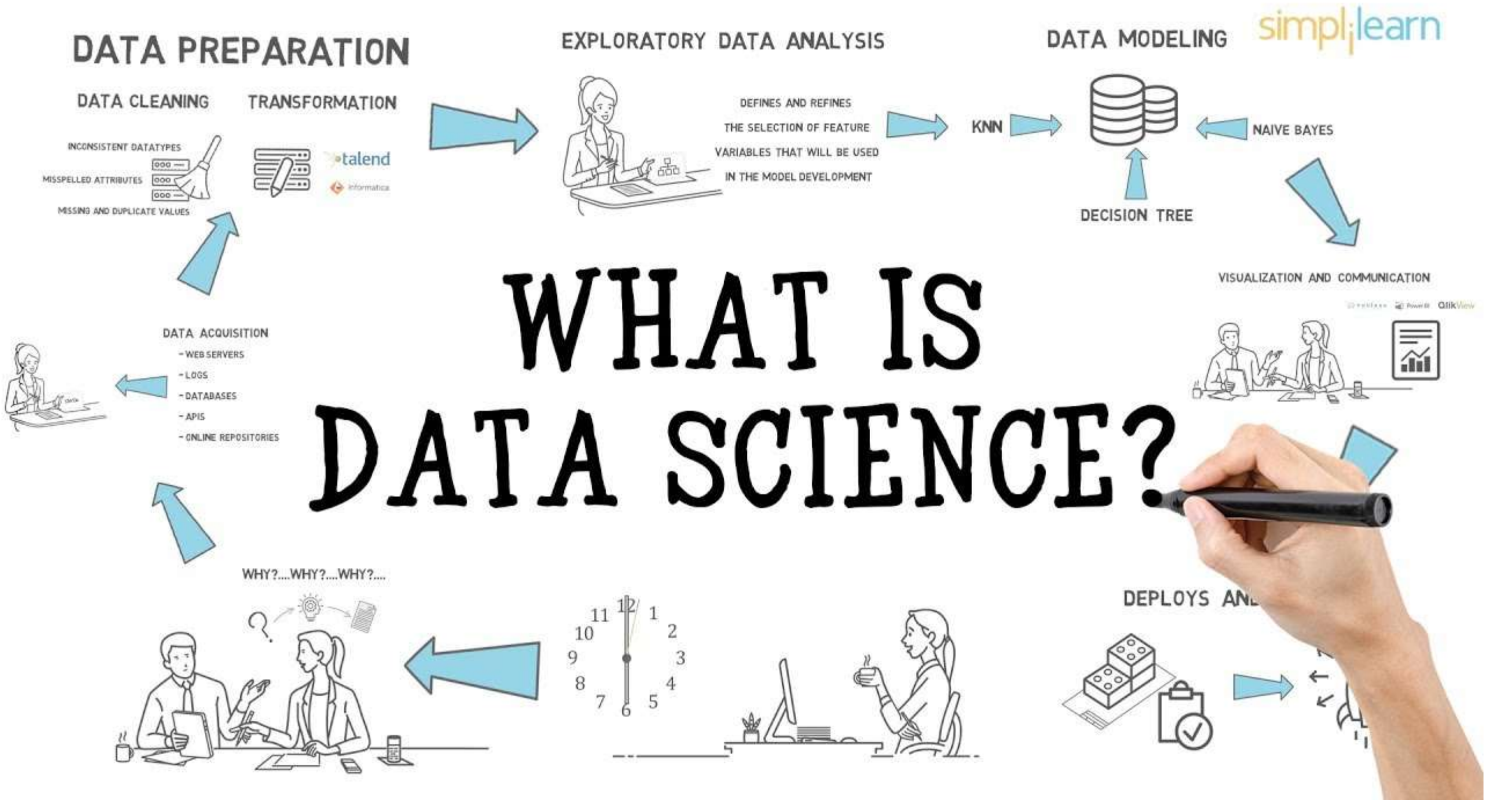
# What makes Data Mining with the Data?




- Refers to the science of collecting meaningful data (past and present) from multiple sources
- Looks for consistent yet unknown patterns and / or relationships between variables.
- Once a new pattern is discovered, you validate the findings by applying the detected patterns to new subsets of data.
- The ultimate goal of data mining is new knowledge (i.e. prediction).
- The 3 most important techniques:
  - Association Analysis
  - Classification
  - Clustering

## What is Data Science?

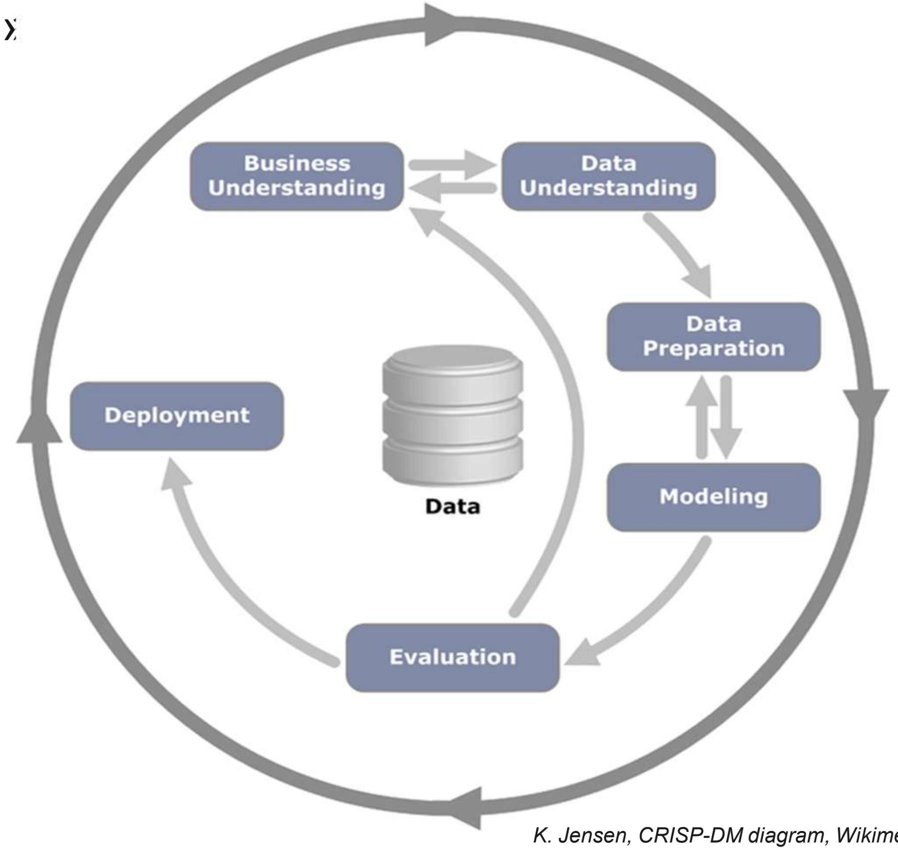
# Data Science in 5 Minutes




https://www.youtube.com/watch?v=X3paOmcrTjQ

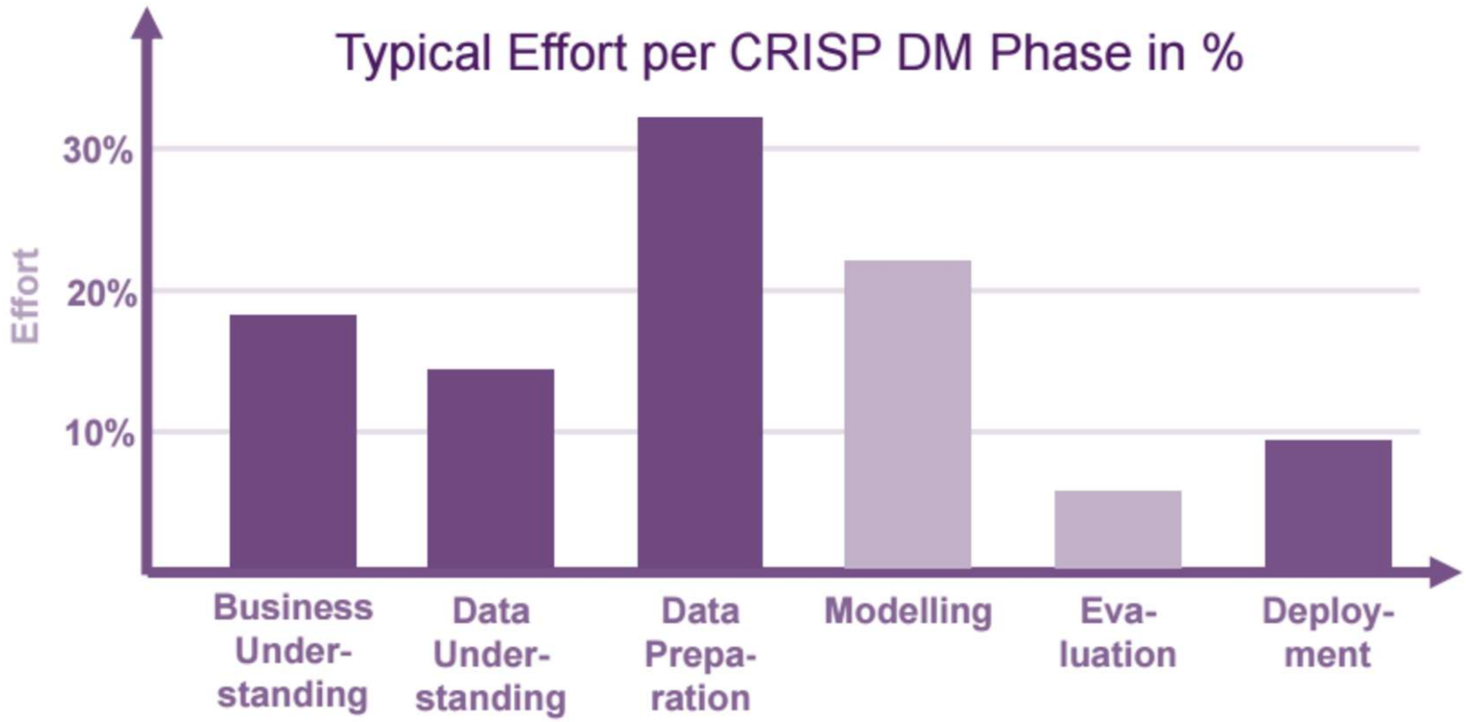
# CRISP-DM: standard process for Data Mining

- «Cross-Industry Standard Process for Data Mining»
  - Leading methodology
- Divided in 6 phases
  - Business Understanding
  - Data Understanding
  - Data Preparation
  - Modeling
  - Evaluation
  - Deployment
- Sequence is not strict
  - iterative with jumps back and forth




*Sources:*
*K. Jensen, CRISP-DM diagram, Wikimedia, 2012*

# Efforts in Data Science




*Sources:*
*Thomas Zeutschler, IT Applications in Business Analytics, Hochschule Düsseldorf*

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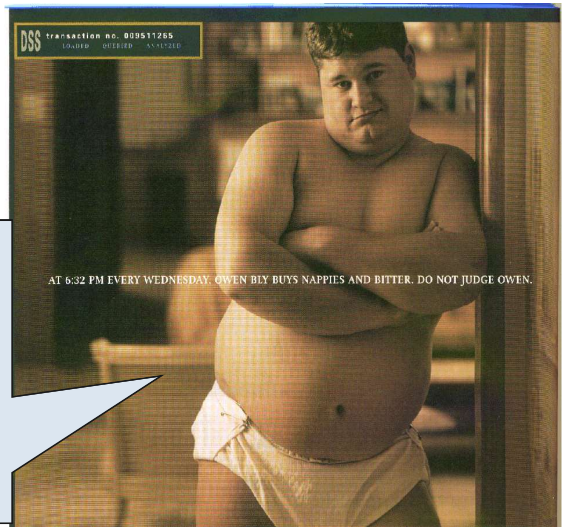
## What kind of patterns can be mined?

# Interessante Fragestellungen aus der Wirtschaft

- Welchen Kunden sollte wann welches Angebot unterbreitet werden?
- Bei welchem Kundenprofil lohnt sich ein Aussendienstbesuch?
- Welche Kunden sind gefährdet?
- Wie hoch ist das Cross-Selling-Potential für ein neues Produkt?
- Welcher Lifetime-Profit läßt sich mit welchem Kunden erzielen?
- Wie lassen sich Interessenten mit hohen Lifetime-Values gewinnen?
- Welcher Umsatz wird im nächsten Jahr erzielt?
- Wie sehen unsere Kunden eigentlich aus?

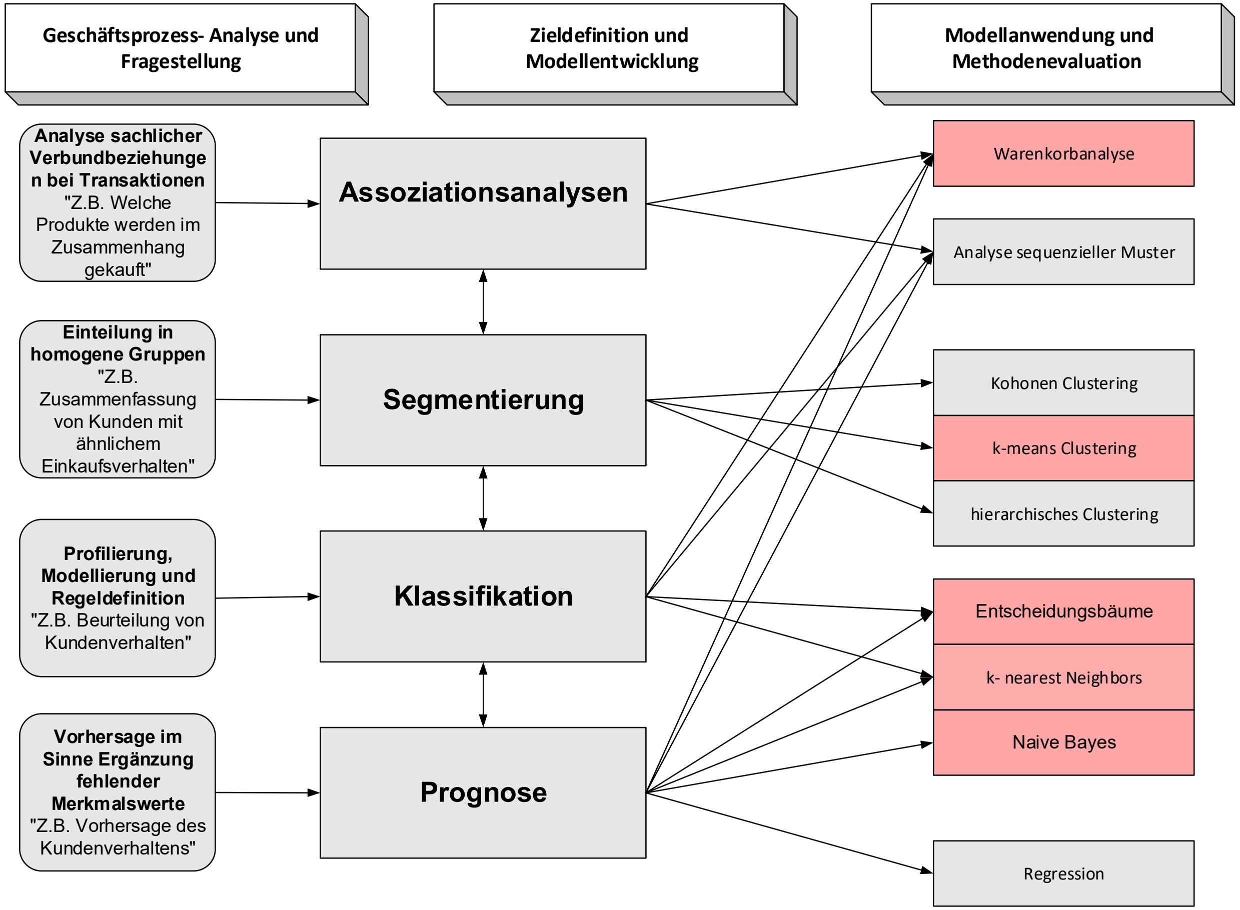
- **Daten sammeln + Nutzen = Wettbewerbsvorteil !!!**

Data Mining-Analysen von POS-Daten ergaben bei einer Handelskette, dass signifikant viele Warenkörbe aus Bier und Windeln bestehen.



## What kind of patterns can be mined?

# From Questions To Methods… an Overview



## What kind of patterns can be mined?

# Data Mining – Association Problems

- What other products are purchased together with a digital camera?
  - Based on previous purchases (shopping cart)
  - E.g., If a digital camera is purchased, flash memory, battery, printer are also purchased.
  - A well-known urban legend, In Walmart, men in their 20s buy beer and diapers together on Fridays.
  - Association Analysis (market basket analysis)

- Similar questions:
  - what items should be displayed together in merchant.
  - What genes appear together in toxic mushrooms.

# Data Mining - Classification Problems

- Is this student going to go to a college?
  - Based on Gender, ParentIncome, ParentEncouragement, IQ, etc.
  - E.g., if ParentEncouragement=Yes and IQ>100, College=Yes
  - Classification (prediction)

- Similar questions:
  - Is this a spam email? (spam filtering)
  - How good/bad is your credit? (credit scoring)
  - Recognition of hand-written letters (pen recognition)
  - What is this gene like? (bioinformatics)
  - Does this person behave like a terrorist?

## What kind of patterns can be mined?

# Data Mining – Segmentation/Clustering Problems

- Who are my Web visitors?
  - Identify similar groups based on demographics, visiting patterns
  - E.g., Daily news readers, email users, shoppers, short-stayers, etc
  - Segmentation (clustering)

- Similar questions:
  - Identify groups of genes (bioinformatics)
  - Identify groups of locations of Cholera incidents in London (spatial data mining)
  - Identify group of customers in merchants (Amazon, E-Bay, MSN, WalMart, BlockBuster, etc) (target marketing)
  - Identify groups of documents. (text categorization)

# Data Mining – Regression Problems

- What is the age of a person?
  - Based on Hobby, MaritalStatus, NumberOfChildren, Income, HouseOwnership, NumberOfCars, …
  - E.g., If MaritalStatus=Yes, Age = 20+4*NumberOfChildren+0.0001*Income+…
  - ➔ Regression (prediction)

- Similar questions:
  - What's the sales amount of ice cream next month? (sales prediction)
  - What's the stock price of MSFT next week? (stock prediction)
  - What's the income of a customer? (marketing)
  - What's the life-time of a software bug? (bug tracking)

What kind of patterns can be mined?

# Data Mining – Anomaly Detection Problems

- Could this network packet be from a virus attack?
  - Predict likelihood of the network packet pattern
  - Anomaly detection (outlier detection)

- Similar questions:
  - Are the hospital lab results normal (Adverse drug effect detection)
  - Is this credit transaction fraudulent? (fraud detection)
  - Does this person behave unusual, maybe worth high-level of security clearance?

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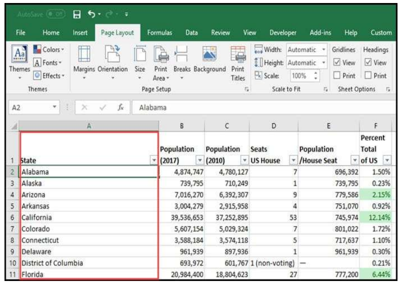
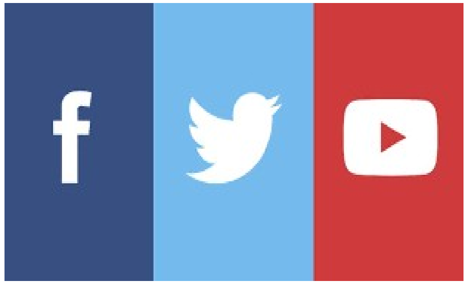
# Different Kinds of Data

- There are many kinds of data that have versatile forms and structures and/or semantic meanings
- Most basic distinction: Structured vs. Unstructured Data



| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | State | Population (2017) | Population (2010) | Seats US House | Population /House Seat | Percent Total of US |
| 2 | Alabama | 4,874,747 | 4,780,127 | 7 | 696,392 | 1.50% |
| 3 | Alaska | 739,795 | 710,249 | 1 | 739,795 | 0.23% |
| 4 | Arizona | 7,016,270 | 6,392,307 | 9 | 779,586 | 2.15% |
| 5 | Arkansas | 3,004,279 | 2,915,958 | 4 | 751,070 | 0.92% |
| 6 | California | 39,536,653 | 37,252,895 | 53 | 745,974 | 12.14% |
| 7 | Colorado | 5,607,154 | 5,029,324 | 7 | 801,022 | 1.72% |
| 8 | Connecticut | 3,588,184 | 3,574,118 | 5 | 717,637 | 1.10% |
| 9 | Delaware | 961,939 | 897,936 | 1 | 961,939 | 0.30% |
| 10 | District of Columbia | 693,972 | 601,767 | 1 (non-voting) | — | 0.21% |
| 11 | Florida | 20,984,400 | 18,804,623 | 27 | 777,200 | 6.44% |


```
118559796:{
  prefName: {
    forename: "Immanuel",
    surname: "Kant"
  },
  variantName: {
    forename: "Emanuele",
    surname: "Kant"
  },
  dateOfDeath: "1804",
  dateOfBirth: "1724",
  profession: "philosopher",
  profession: "professor",
  profession: "librarian"
}
```

Modulbeschreibung: Data Science

**Lernziele:**

**Fachkompetenzen:**
Die Teilnehmenden können:
- Ausgewählte Themen von Informationssystem- und Datenbanksystem-Anwendungen kennen und die entsprechenden Werkzeuge anwenden können.
- Sie kennen den Data-Mining (DM) Prozess und verschiedene DM-Anwendungen. Sie kennen die Eigenschaften der wichtigsten Algorithmen für die Klassifikation, die Segmentierung und für die Warenkorbanalyse und können diese an einfacheren Szenarien anwenden
- Sie werten Daten aus und erstellen Datengrundlagen für Entscheidungen.
- Sie wenden Daten als Basis für Prozessverbesserungen, neue Produkte, Services und darauf zugeschnittene Geschäftsmodelle an.

**Methodenkompetenzen:**
Die Teilnehmenden können:
- Grundlagen der Datenkonsolidierung und Datenaufbereitung
- Grundlagen Data Mining
- Grundlagen im Umgang mit Jupyter / Python

**Selbstkompetenzen:**
Die Teilnehmenden können:
- Selbstmotivation an einem schwierigen Thema

- There are different scales of measurement each with different properties
  - nominal, ordinal, interval, ratio
- There are also special forms of data
  - Data streams, graph/networked data, spatial data, multimedia data, websites

# Data Sets and Data Objects

- Data Sets are made up of Data Objects
- A Data Object represents an entity
  - e.g. a customer, a store item, a patient, a student, a course
  - Other names for data objects are: sample, instance, data point, object, data tuples
- Data Objects are typically describes by one or more Attributes
  - Fields that represents a characteristic of a data object
  - Other names for attributes are: dimension, feature, variable
  - Observed values for an attribute are called observations
  - A set of attributes used to describe an object is called attribute vector or feature vector
  - Analysis of that with only one, two or more attributes are called univariate, bivariate, multivariate
  - There are several types of attributes: nominal (also: binary), ordinal and cardinal (interval / ratio)

# Types of Attributes (1/2)

- **Nominal** (also known as categorical)
  - The values of a nominal attribute relate to names or things without any meaningful order
  - E.g. hair colors such as black, brown, blond, red, gray and white
  - It’s referred to as a Binary Attribute, if there are only two possible attribute values
- **Ordinal** (also known as categories)
  - Refers to an attribute with possible values that have a meaningful order to rank among them
  - E.g. drink sizes such as tall, large and venti
- **Cardinal** (also known as numeric)
  - An attribute value of «0» is different to a real zero-point
  - **Interval-scaled** attributes allows us to compare and quantify the diference between values
  - **Ratio-scaled** attributes allows us to speak about a value as being a multiple (or ratio) of another

# Types of Attributes (2/2)

| Scale | True Zero | Equal Intervals | Order | Category | Example |
|---|---|---|---|---|---|
| Nominal | No | No | No | Yes | Marital Status, Sex, Gender, Ethnicity |
| Ordinal | No | No | Yes | Yes | Student Letter Grade, NFL Team Rankings |
| Interval | No | Yes | Yes | Yes | Temperature in Fahrenheit, SAT Scores, IQ, Year |
| Ratio | Yes | Yes | Yes | Yes | Age, Height, Weight |

# A Sample Dataset

- Outlook, Temp Humidity, Windy and Play Golf are Attributes
  - Outlook is Nominal
  - Play Golf is Binary
  - Windy is Boolean
  - Humidity is Ordinal
  - Temp is Cardinal
- An instance or data object in the set is

| | | | | |
|---|---|---|---|---|
| Rainy | 28.3 | High | False | No |

  which also forms a feature vector.
- Rainy, Overcast and Sunny are observations of the attribute Outlook

| Outlook | Temp | Humidity | Windy | Play Golf |
|---|---|---|---|---|
| Rainy | 28.3 | High | False | No |
| Rainy | 27.2 | High | True | No |
| Overcast | 27.4 | High | False | Yes |
| Sunny | 22.7 | High | False | Yes |
| Sunny | 14.3 | Normal | False | Yes |
| Sunny | 11.9 | Normal | True | No |
| Overcast | 13.8 | Normal | True | Yes |
| Rainy | 19.8 | High | False | No |
| Rainy | 9.6 | Normal | False | Yes |
| Sunny | 21.1 | Normal | False | Yes |
| Rainy | 20.8 | Normal | True | Yes |
| Overcast | 19.8 | High | True | Yes |
| Overcast | 25.3 | Normal | False | Yes |
| Sunny | 22.8 | High | True | No |

## What is Data like?

# Always - Get to Know your Data !!!

- Before direcly jumping into Data Mining one should get familiar with its data
  - It may be noisy, enormous in volume and originate from heterogenous sources
  - A closer look at the data will help to ease the preprocessing, which is one major task in data mining
- Relevant Questions
  - What kind of attributes/fields make up my data?
  - What kind of values does each attribute have?
  - Which attributes are discrete and which are continuous?
  - How are the values distributed?
  - Are there ways to visualize the data to get a better sense of it all?
  - Are there any outliers?

# Some Major Issues in Data Mining

- Wide spectrum of tasks makes it a dynamic and fast-growing research field with numerous methods
- Searches patterns among combinations of dimensions at various levels of abstraction (Multidimensional)
- Depending on the task methods from other disciplines must be integrated (NLP, IR)
- Uncertainty, Incompleteness and Noise may lead to erroneous patterns
- Not all found patterns are of interest
- Presentation and visualization of the results that can be easily understood
- Huge amount of data must be processed (sometimes in real-time)
- Handling the complexity and diversity of data from many different sources
- Violation of individual privacy and potential misuse
- Quality of the data (see next slide as an example)

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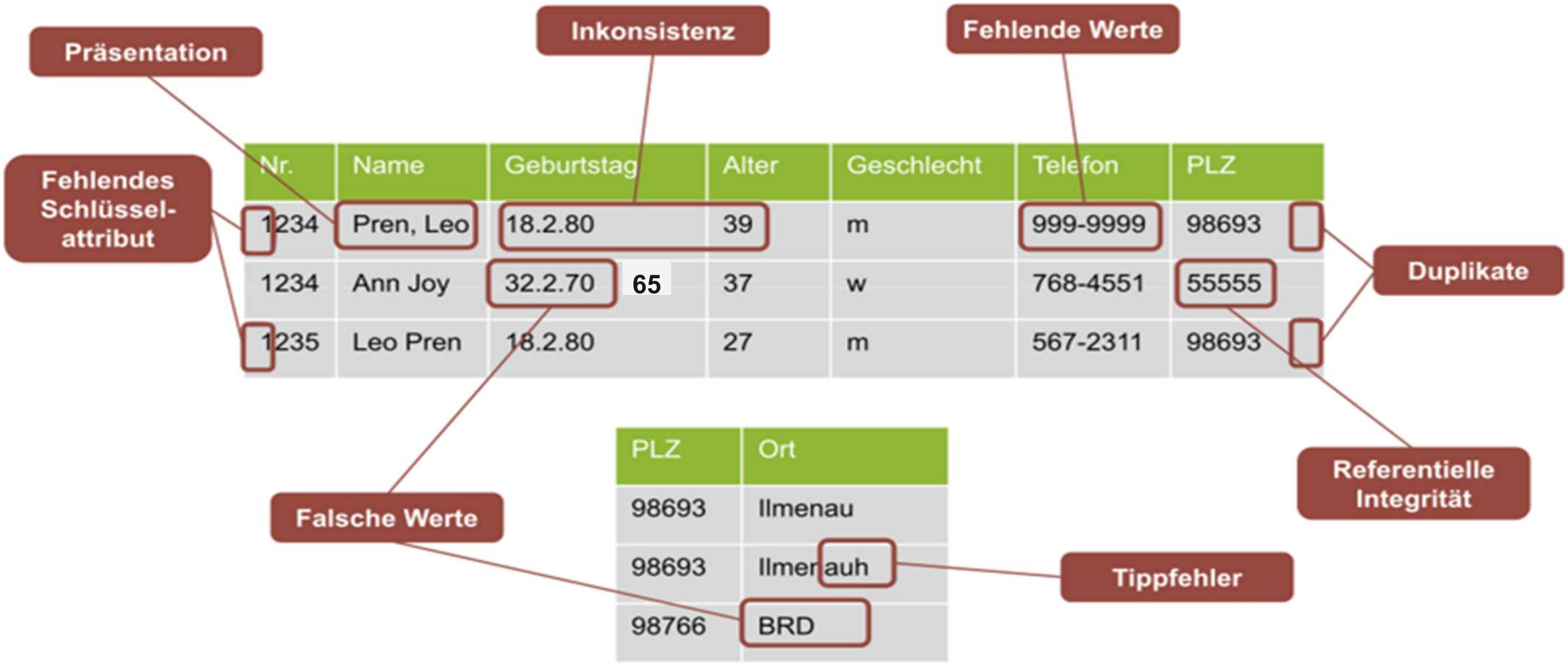
# Quality of the Data is an Major Issue



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# Data Quality is defined by several Elements

| Element | Example |
|---|---|
| Accuracy | |
| Completeness | |
| Consistency | |
| Timeliness | |
| Believability | |
| Interpretability | |
| (Value Adding) | |
| (Acessibility) | |

# Motivation for Preprocessing

**«Low quality data will lead to low quality mining results»**

- Several techniques work together to improve the quality of the data
  - Data Integration: integration of data from multiple sources
  - Data Cleaning: fill in missing values, remove noise and outliers, correct inconsistencies
  - Data reduction: to reduce size of the representation by removing irrelevant featues
  - Data transformation: better scaling to ranges or attribute types

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# Measuring Central Tendencies (1/2)

- Mean
  - Most common and very effective numeric measure of the «center»
  - Let $x_1, x_2, x_3, \ldots, x_N$ be a set of $N$ observations of an attribute $X$ then the mean is:

$$\bar{x} = \frac{\sum_{i=1}^{N} x_i}{N}$$

  - If there is a weight $w_i$ assigned for each value $x_i$ the so called weighted arithmetic mean or weighted average can be computed:

$$\bar{x} = \frac{\sum_{i=1}^{N} w_i x_i}{\sum_{1}^{N} w_i}$$

  - A substantial drawback is that the mean is sensitive to extremes (outliers)
  - A workaround is to remove some of the top and bottom values → trimmed mean

| Temp |
|---|
| 28.3 |
| 27.2 |
| 27.4 |
| 22.7 |
| 14.3 |
| 11.9 |
| 13.8 |
| 19.8 |
| 9.6 |
| 21.1 |
| 20.8 |
| 19.8 |
| 25.3 |
| 22.8 |

$\bar{x} = 20.34$

$N = 14$

# Measuring Central Tendencies (2/2)

- Median
  - Middle value in a set of ordered values
  - Separates the higher half from the lower half
  - More robust against outliers and extreme values i.e. when the data is skewed
  - If the set has an odd number of values then the median is unique
    → simply the middle value of the set
  - If the set has an even number of values then the median is not unique
    → it is the two middlemost values and any value in between
    → for numeric values the median is taken as the average
- Mode
  - The value that occurs most frequently in the set
  - There may be more than one mode (unimodal, bimodal… multimodal)

| Temp |
|---|
| 28.3 |
| 27.2 |
| 27.4 |
| 22.7 |
| 14.3 |
| 11.9 |
| 13.8 |
| 19.8 |
| 9.6 |
| 21.1 |
| 20.8 |
| 19.8 |
| 25.3 |
| 22.8 |

| Temp |
|---|
| 9.6 |
| 11.9 |
| 13.8 |
| 14.3 |
| 19.8 |
| 19.8 |
| 20.8 |
| 21.1 |
| 22.7 |
| 22.8 |
| 25.3 |
| 27.2 |
| 27.4 |
| 28.3 |

$$median = \frac{20.8+21.1}{2} = 20.95$$

$$mode = 19.8 \quad N = 14$$

# Measuring the Dispersion (1/3)

- Range
  - The difference between the largest and the smallest value:
  - $R = \max(X) - \min(X)$
  - Can be misleading when there are extremely high or low values
- Quantiles
  - $q$-quantiles are $q-1$ data points $Q_k$ that split the data set in $q$ equal-size sets
  - If you have an uneven set of numbers, you have to calculate the middle.
  - The 4-quantiles are known as quartiles and the 100-quantiles as percentiles
- Interquartile Range (IQR)
  - Distance between 1st and 3rd quartile is a simple measure of the spread:
  - $IQR = Q_1 - Q_3$

| Temp |
| --- |
| 9.6 |
| 11.9 |
| 13.8 |
| 14.3 |
| 19.8 |
| 19.8 |
| 20.8 |
| 21.1 |
| 22.7 |
| 22.8 |
| 25.3 |
| 27.2 |
| 27.4 |
| 28.3 |

$Q_1 = 14.3$

$Q_2 = 20.95$

$Q_3 = 25.3$

$Range = 18.7$

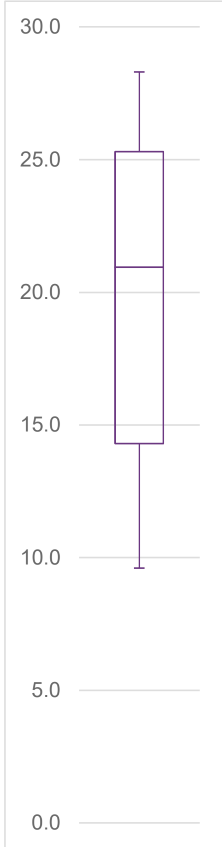
$IQR = 11$

$N = 14$

# Measuring the Dispersion (2/3)

- Five Number Summary
  - No single numeric measure of spread is very useful for describing skewed distributions
  - But still the quantiles and the median lacks information about the endpoints of a distribution
  - Thus the five number summary consists of: $Minimum, Q_1, Median, Q_3, Maximum$
- Boxplots
  - A graphical representation of the five number summary as follows:
  - Q1 and Q3 mark the ends of the box – IQR gives ist length
  - The median is marked as a line within the box
  - Two lines (whiskers) extend the box to the smallest and largest observations (as long as they're within 1.5 x IQR of the quartiles)
  - Outliers may be visualized individually

| Temp |
|---|
| 28.3 |
| 27.2 |
| 27.4 |
| 22.7 |
| 14.3 |
| 11.9 |
| 13.8 |
| 19.8 |
| 9.6 |
| 21.1 |
| 20.8 |
| 19.8 |
| 25.3 |
| 22.8 |



# Measuring the Dispersion (3/3)

- Standard Deviation
  - A low standard deviation means that the data observations tend to be close to the mean
  - A high standard deviation means that teh data observations are spread out over a large range of values
  - The standard deviation $\sigma$ is the square root of the variance $\sigma^2$
  - About 68% of all values are no more than $\sigma$ away from the mean
- Variance
  - The variance is the average of the squared differences from the mean
  - Let $x_1, x_2, x_3, \dots, x_N$ be a set of $N$ observations and $\bar{x}$ the mean of attribute $X$ then the variance is:

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2$$

| Temp |
|---|
| 28.3 |
| 27.2 |
| 27.4 |
| 22.7 |
| 14.3 |
| 11.9 |
| 13.8 |
| 19.8 |
| 9.6 |
| 21.1 |
| 20.8 |
| 19.8 |
| 25.3 |
| 22.8 |

$\bar{x} = 20.34$

$\sigma^2 = 5.97$

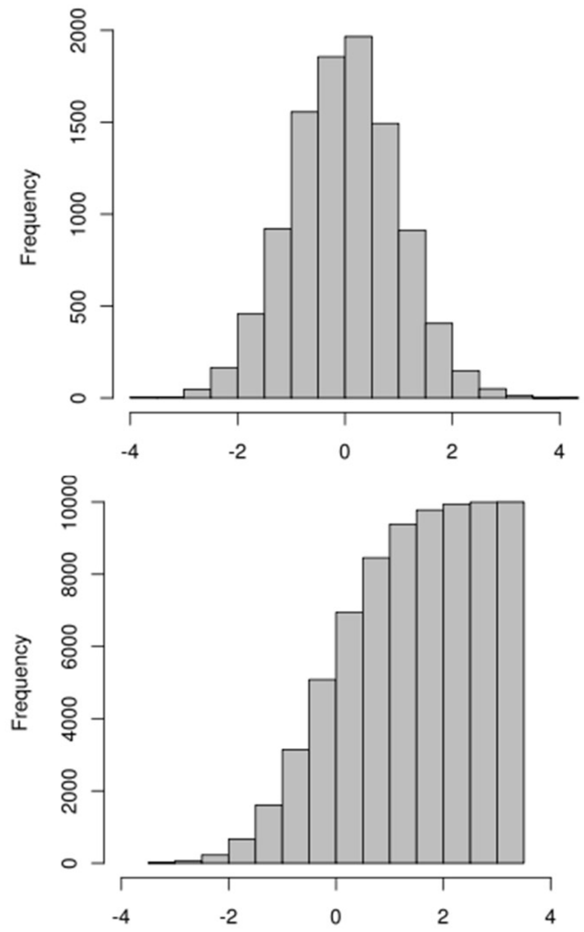
## Agenda

# What we will cover today…

- Administratives / Course Overview
- What is Data Science?
- What kind of patterns can be mined?
- What is Data like?
- Data Quality and Preprocessing Motivation
- Basic Statistical Description of Data
- **Rough Data Visualization Overview**

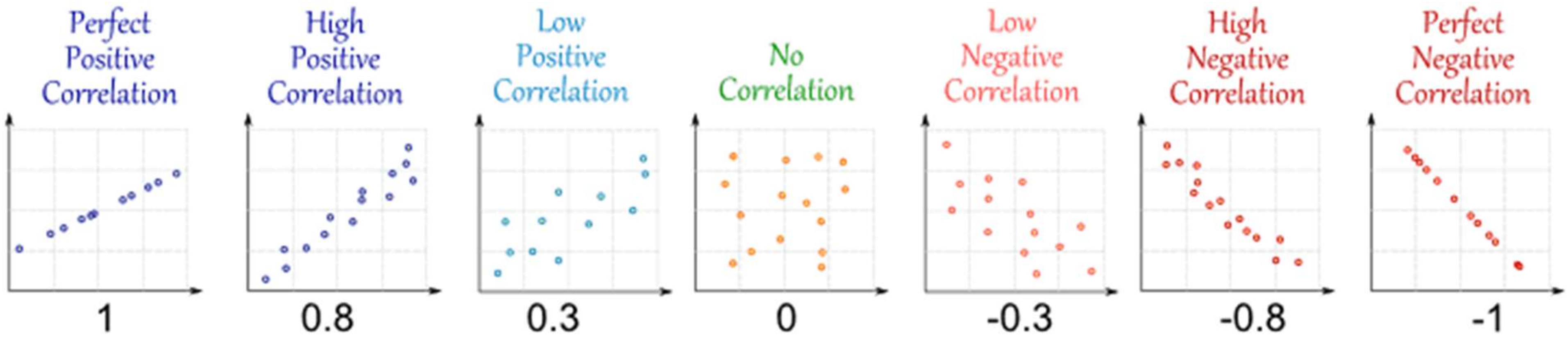
# Histograms

- A histogram is an approximate representation of the distribution of numerical or categorical data of one attribute.
  - Very widely used - for nominal data it's also known as bar chart
  - Give a sense of the density of the underlying distribution of the data
- For nominal attributes a vertical bar for each known value of the attribute is drawn.
  - The height indicates the frequency or count of that value.
  - Sometimes there are gaps between the bars
- For cardinal attributes the range of values are partitioned into disjoint consecutive subranges (also known as bins)
  - Typically the bins have the same width (equiwidth-binning)
  - The height indicates the frequency of observed items within that subrange



# Scatter Plots

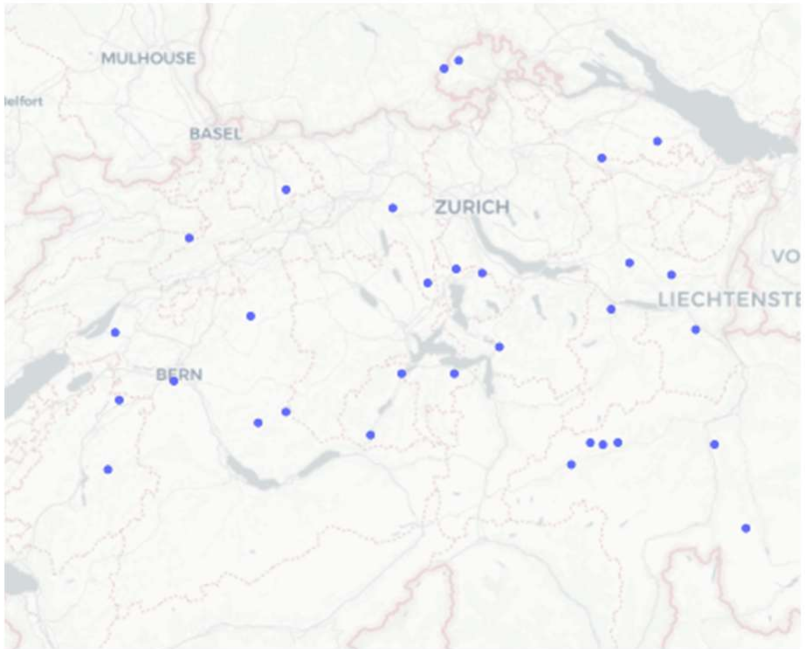
- Type of plot using Cartesian coordinates to display values for typically two attributes.
  - Data is displayed as a collection of points, each having the value of one variable determining the position on the horizontal axis and the value of the other variable determining the position on the vertical axis
  - Visualization is possible for all attribute types, even when mixed
- Easy to determine if there appears to be a relationship, pattern or trend between numeric attributes
  - Very useful for a first look, to see clusters, outliers

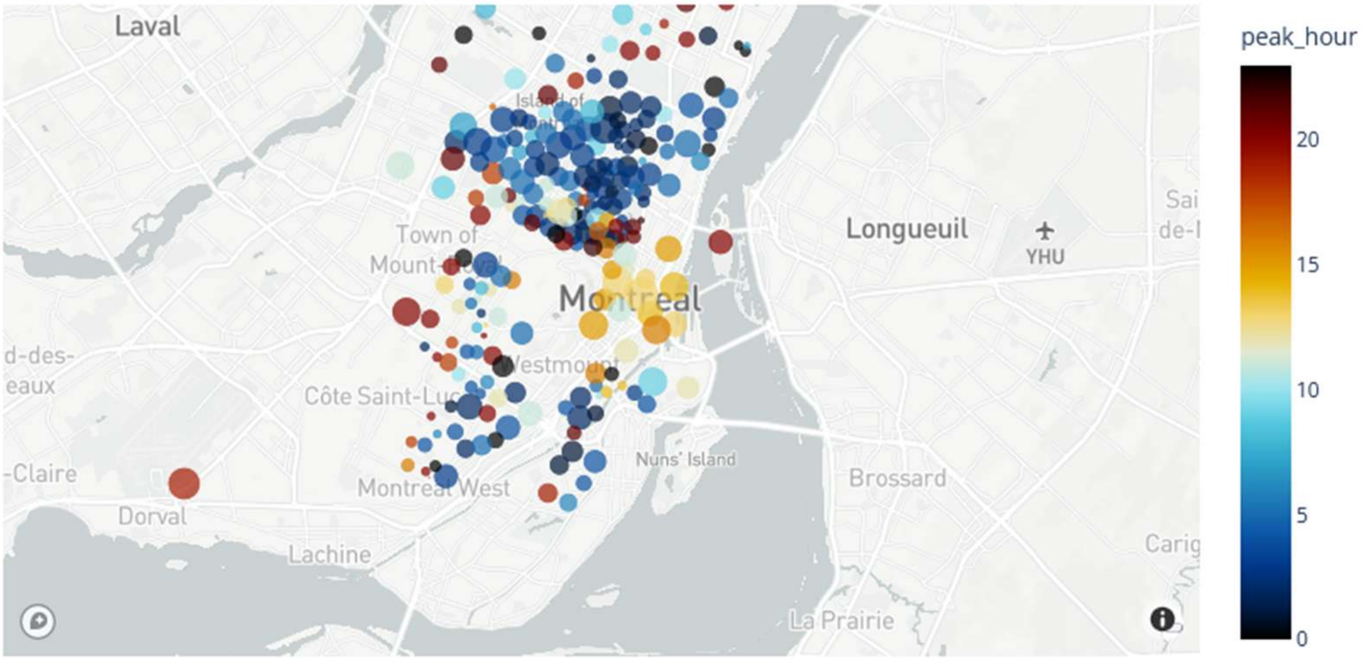


## Rough Data Visualization Overview

# Scatter Plots on Mapbox

- Plotly/Mapbox allows to visualize data on a map
  - Many different styles available, see: https://plotly.com/python/mapbox-layers/



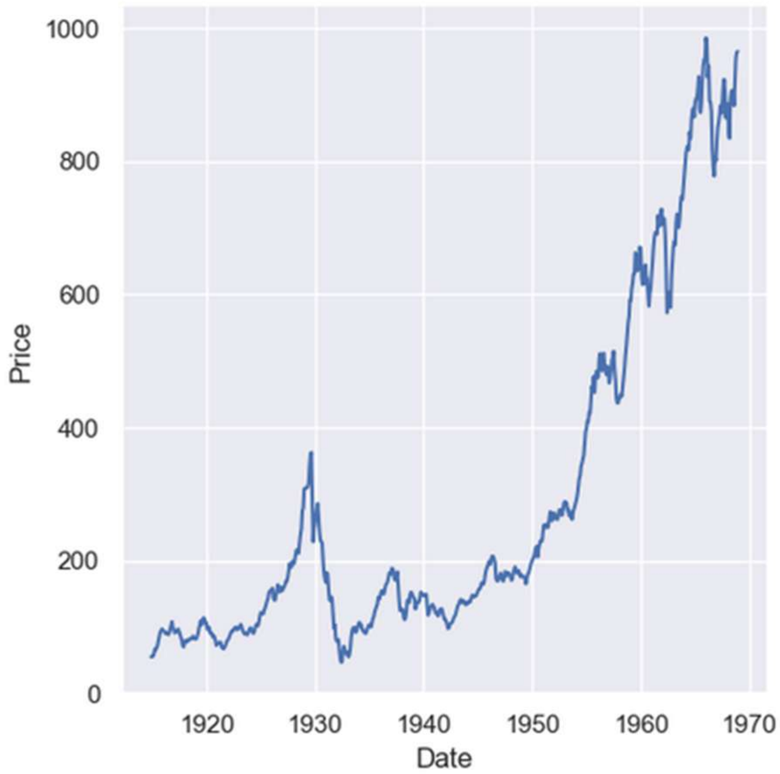



# Line Graphs

- A line plot is a relational data visualization showing how one continuous variable changes when another does.
- It's one of the most common graphs widely used in finance, sales, marketing, healthcare, natural sciences, and more.

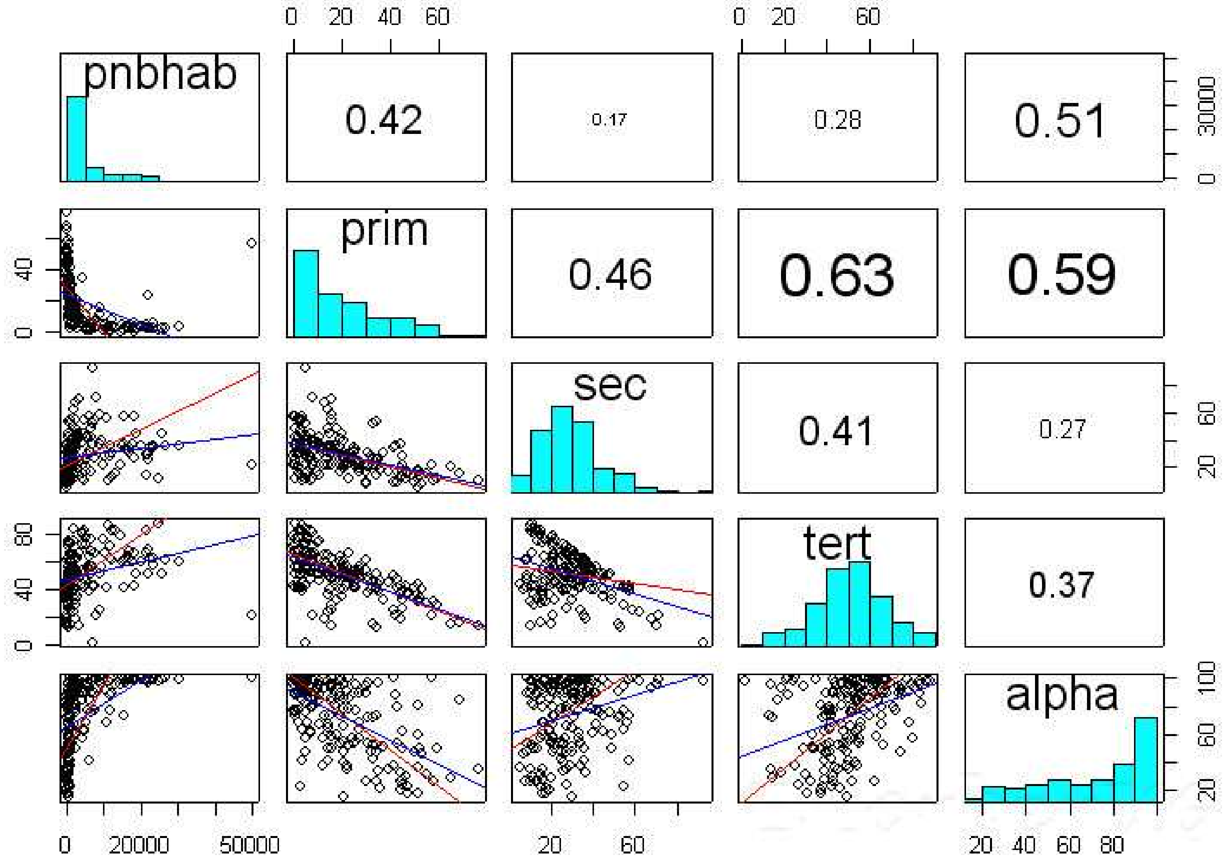
| | Date | Price |
|---|---|---|
| 0 | 1914-12-01 | 55.00 |
| 1 | 1915-01-01 | 56.55 |
| 2 | 1915-02-01 | 56.00 |
| 3 | 1915-03-01 | 58.30 |
| 4 | 1915-04-01 | 66.45 |

```
sns.relplot(
    data=dow,
    x="Date",
    y="Price",
    kind="line",
    estimator="mean",
    errorbar="ci",
    dashes=False,
    markers=True
)
```



# Geometric Projections of Multidimensional Data

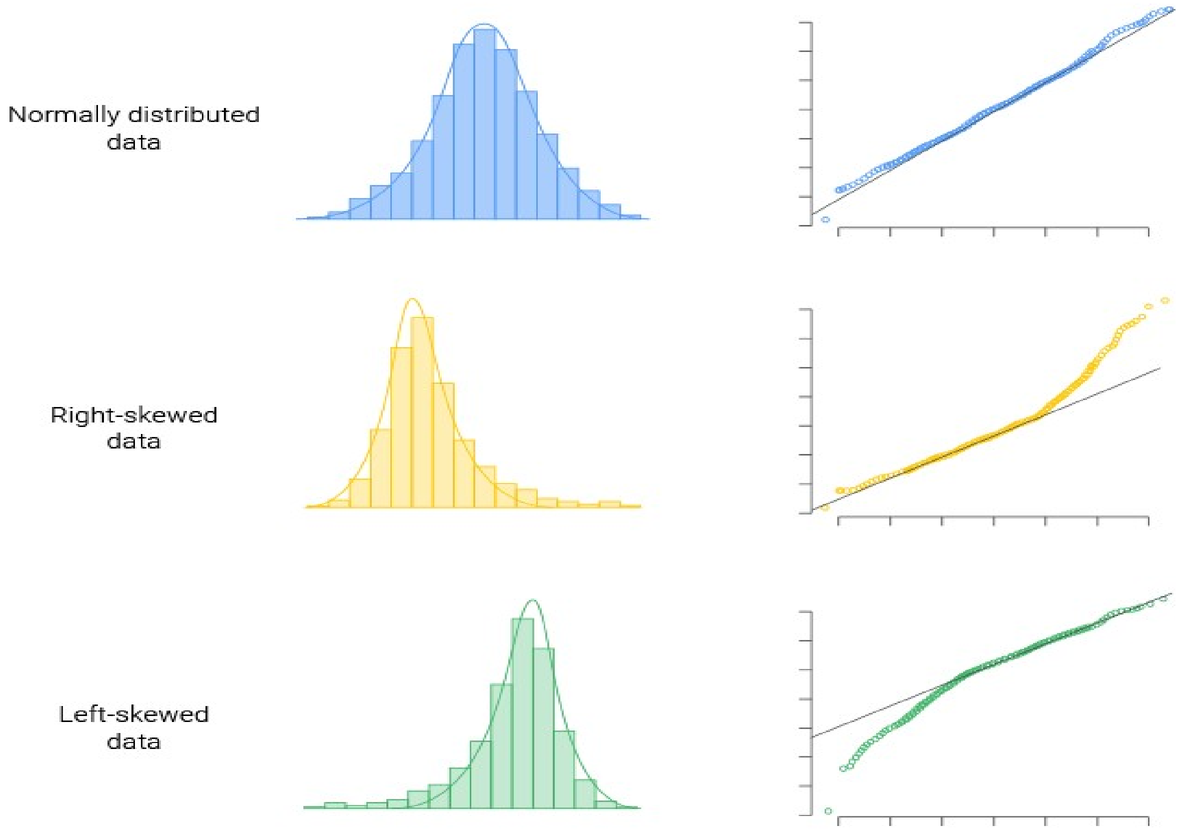
- Challenge: How to visualize a multidimensional space in 2-D?
- Scatterplot matrix
  - Extension to scatter plots
  - Creates an $n \times n$ Matrix of plots for an $n$-dimensional data set
  - Becomes less effective as the dimensionality increases
- Lines of best fit (or trends)
- Histograms in the diagonal
- Correlation coefficients in the upper part of the matrix.




Eugene Horber, University of Geneva, 2019

# Q-Q-Plots

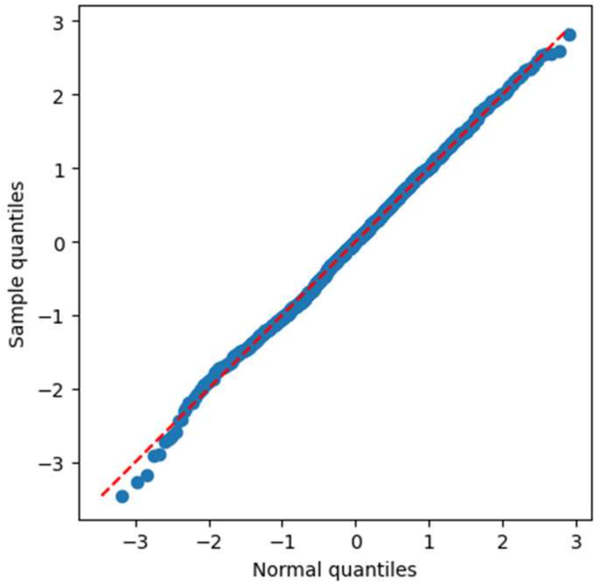
- Sometimes it's important to know whether the data is normally distributed.
- A quantile-quantile plot (QQ plot) is a good first check.
- It shows the distribution of the data against the expected normal distribution.
- If the data is normally distributed, the points fall on the 45° reference line. If the data is non-normal, the points deviate noticeably from the reference line.



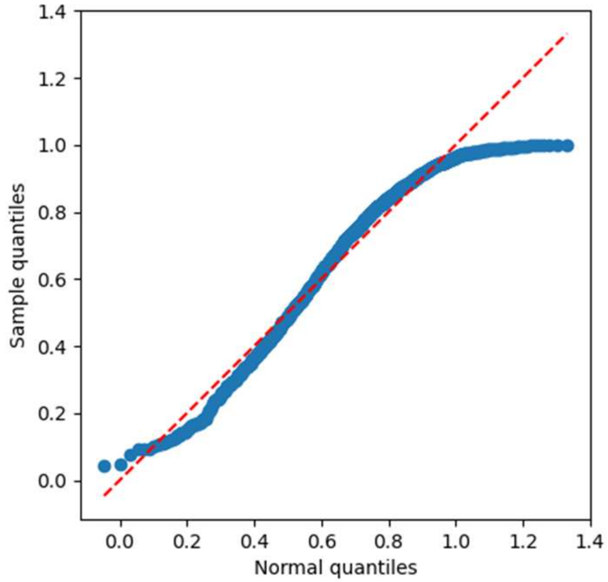

https://www.learnbyexample.org/r-quantile-quantile-qq-plot-base-graph/, 2023

# Sample Q-Q-Plots

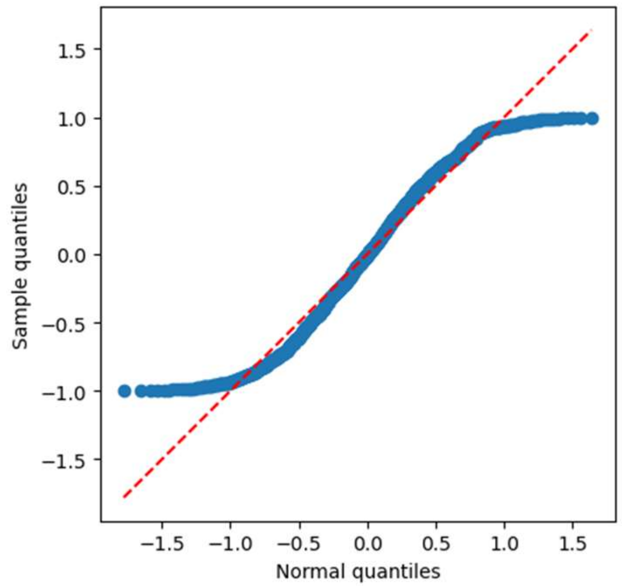
```
# Make up some normal distributed data
x = np.random.normal(loc=0.0, scale=1, size=1000)
QQ_plot(x)
```




```
# Make up some x-square distributed data
x = np.random.power(2, 1000)
QQ_plot(x)
```

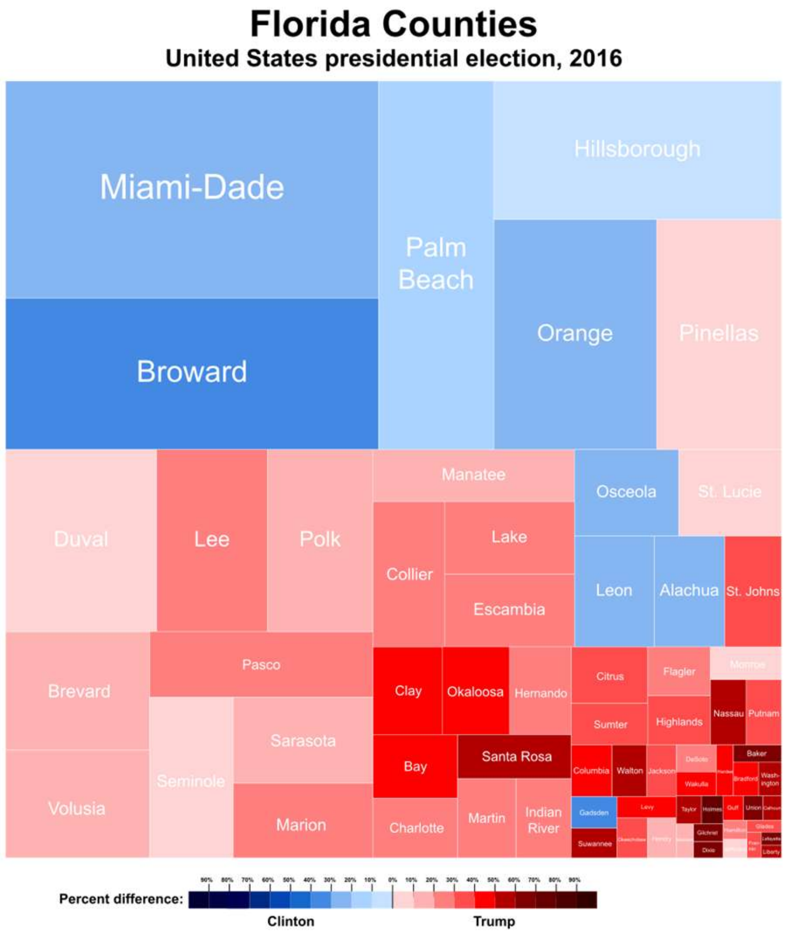



```
# Make up some dummy uniform distributed data
x = np.random.uniform(-1, 1, 1000)
QQ_plot(x)
```



# Tree Maps

- Treemaps display data as a set of nested rectangles.
  - Each branch of the tree is given a rectangle, which is then tiled with smaller rectangles representing sub-branches.
  - A leaf node's rectangle has an area proportional to a specified dimension of the data
  - Often the leaf nodes are colored to show a separate dimension of the data.
- Hierarchical approach
  - Instead of visualize multiple dimensions simultaneously, the dimensions are partitioned into subsets
  - Allows a drill-down to see more



# Excursus: Tag Clouds

- A tag cloud is a novelty visual representation of text data
  - Also known as word cloud
  - used to depict keyword metadata (tags) on websites, or to visualize free form text.
- Tags are usually single words
  - The importance of each tag is shown with font size or color
- Different applications
  - Frequency, Significance, Category
- This format is useful for quickly perceiving the most prominent terms to determine its relative prominence.


*Spathis, D., Linguistic Patternc in best-selling books, 2015*

# Vielen Dank!